

OBJECTIVE

Gold-medalist Software Engineering graduate with hands-on research in computer vision, vision-language models, and medical imaging. I aim to make large vision models faster, lighter, and more trustworthy — through token compression, knowledge distillation, and interpretable deep learning — so they can run reliably in real-world and resource-limited settings.

RESEARCH INTERESTS

Efficient Deep Learning · Interpretable AI for Medical Imaging · Computer Vision · Self-Supervised Representation Learning · Image Recognition & Object Detection · Vision-Language Models

MAJOR RESEARCH PROJECT

2025 – 2026

Augmented Reality & Deep Learning Assisted Enhanced Museum Experience

Final Year Research Project (Thesis) — University of Haripur

- Built a multimodal AI system integrating computer vision, deep learning, speech interaction, and multilingual assistance for real-world cultural heritage recognition (case study: Taxila Museum).
- Curated a custom dataset of Taxila Museum artifacts; performed end-to-end data collection, preprocessing, and augmentation pipelines; developed vision-language recognition pipelines for low-resource artifact categories.
- Research paper on hybrid vision-language recognition for cultural heritage systems under preparation for publication.
- Nominated by the Directorate General of Science & Technology (DoST), KP for research funding, recognizing scientific merit and societal impact.
- Awarded Top Project at the Final Year Project Exhibition, University of Haripur.

ACADEMIC PROJECTS

PulmoVision: Automated Pneumonia Diagnosis Using Deep Learning — Python, TensorFlow, ResNet

- ResNet transfer learning pipeline for pneumonia classification from chest X-rays, deployed as a public web app on Streamlit Community Cloud.

[Live App](#) · [Code](#)

Performance Benchmarking of Machine Learning Algorithms — Python, scikit-learn, Pandas

- Benchmarking multiple machine learning algorithms across datasets and problem instances, analyzing performance variations to identify relationships between dataset characteristics and algorithm effectiveness.

NailScan: AI-Powered Nail Condition Analyzer — Python, PyTorch, YOLOv5, OpenCV

- Built and deployed a nail health screening app: a YOLOv5 segmentation model locates the nail, then a ResNet-152 classifier grades it into one of 6 conditions.

[Live App](#) · [Code](#)

Encoder-Aware Visual Token Compression for Efficient VLMs — Python, PyTorch, ViT

- Implementing adaptive, encoder-dependent visual token compression for vision-language models, benchmarking a random baseline and Top-K pruning against an adaptive compression method on ViT.

[Code](#)

Data Structures & Algorithms Visualization Suite — C++

- Desktop application with graphical visualization of sorting, tree traversals, and graph pathfinding algorithms.

TECHNICAL SKILLS

Languages	Python, JavaScript, C++
Frameworks	PyTorch, TensorFlow, Keras, Hugging Face Transformers, OpenCV, Scikit-learn, FastAPI, Pandas, Seaborn, Plotly, Streamlit
Dev Tools	Git/GitHub, MySQL, SQL, Linux/Bash, REST APIs, OOP, Web Development

EXPERIENCE

Jul 2026 – Present	AI Intern <i>UAVs Lab, National Radio and Telecommunication Corporation (NRTC) — Haripur, Pakistan</i> - Working on an object detection and pattern recognition module for a UAV-based mission: comparing before/after aerial imagery from a drone's outbound and return flight path to identify and flag missing objects. - Collaborating within a larger multi-module team project, owning the detection and comparison component end-to-end.
Jul 2025 – Aug 2025	IT Directorate Intern <i>Senate of Pakistan — Islamabad, Pakistan</i> - Contributed to software development and web-based systems within the IT Directorate. - Assisted in maintaining and troubleshooting internal web applications used across departments. - Collaborated with senior IT staff on requirement gathering and testing for ongoing system upgrades.
Jul 2024 – Aug 2024	Game Development Intern <i>Codematics Inc. — Abbottabad, Pakistan</i> - Developed 2D/3D Unity projects, including animations and shader graphs. - Explored AI-driven gameplay elements and prototyped interactive mechanics. - Collaborated with a small team on game design iteration and playtesting feedback.

EDUCATION

Oct 2022 – Jun 2026	B.S. Software Engineering — Gold Medalist, CGPA 3.92/4.00 <i>University of Haripur, KP, Pakistan</i> <i>Thesis: Augmented Reality & Deep Learning Assisted Enhanced Museum Experience</i>
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AWARDS & ACHIEVEMENTS

Gold Medal, BS Software Engineering (CGPA 3.92/4.00), University of Haripur — Jun 2026
Top Project, Final Year Project Exhibition, University of Haripur — Jul 2026
Laptop Recipient, PM Youth Program (Academic Excellence) — Nov 2023
Top Performer, Python Frenzy Competition, IT Club — Sep 2024
A1 grades in SSC & FSc (2019–2021); Top 10, Abbottabad Board (FSc); 93rd percentile, Engineering/IT entry test

CONFERENCES & SEMINARS

AI in Healthcare Webinar — Abu Dhabi, UAE
1st International Conference on Computational Sciences and Innovations (ICCSI) — Haripur, Pakistan
CyberFest — Peshawar, Pakistan

CERTIFICATIONS

- AI in Healthcare Specialization — Stanford University
- Deep Learning for Computer Vision: Techniques and Applications — Khalifa University
- Computer Vision Specialization — University of Colorado Boulder
- Modern AI Models for Vision and Multimodal Understanding — University of Colorado Boulder
- Multimodal Intelligence: Vision, Audio and Language in Action — Coursera
- Unsupervised Learning, Recommenders, Reinforcement Learning — DeepLearning.AI
- Fundamentals of Building AI Agents — IBM
- Explainable AI: Scene Classification and Grad-CAM Visualization — Coursera
- Interpretable Machine Learning — Duke University
- Build Multimodal Generative AI Applications — IBM
- AI in Healthcare Webinar — UAEU
- Crash Course on Python — Google

LEADERSHIP & SERVICE

Jun 2025 – Present	Vice President, IT Computing Society, University of Haripur
Dec 2025	Career Expo 2025 Organizer, University of Haripur [Certificate]
Apr 2025	Event Manager, 4th Convocation, University of Haripur [Certificate]
Jan 2024	Team Lead, Social Work Visit — Aghosh AI-Khidmat, Haripur [Photos]